

Department of Computer Science and
Engineering,
School of Technology,
Pandit Deendayal Energy University
(PDEU)
Software Engineering(24CS209P)

Lab Experiment - 1

Name: Dax B Patel
Roll no: 24BCP269

Project Abstract

“The Library Management System is a web-based application designed to automate and streamline library operations, including book management, member records, and transactions. The system enables efficient cataloguing, real-time availability tracking, and secure issuance and return of books.”

Questionnaire

- What types of users should the system support ?
- What permissions should each user role have ?
- Should users be able to register themselves, or should accounts be created only by the administrator?
- How should books be categorized ?
- Do you want to track **individual physical copies** of each book or only the **total number of available copies**?
- Should the system support digital resources such as e-books, PDFs, or journals?
- Do you require advanced search options ?
- Should the system automatically calculate late return fines?
- What should be the fine amount per day for overdue books?
- Are there any exemptions from fines ?
- Should users be able to reserve or place a hold on books that are currently unavailable?
- How long should a reserved book remain blocked once it becomes available?
- Should notifications be sent when a reserved book becomes available?
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Use case:

- Issue Book

Scope:

- The scope of this use case is the **Library Management System**, which allows librarians to issue books to registered users.

Description:

- This use case describes how a librarian issues a book to a student using the library system.

Flow Description:

- This use case explains the interaction between the **librarian** and the **system** to successfully issue a book to a student.

Precondition:

- The librarian is logged into the system.
- The student is a registered library member.
- The requested book exists in the library database.

Activation:

“This use case starts when the librarian selects the **Issue Book** option from the system menu.”

Main flow:

1. Librarian enters the student ID in the system.
2. System validates the student's account.
3. Librarian enters or scans the book ID.
4. System checks the availability of the book.
5. System assigns the book to the student.
6. System updates the book status as **Issued**.
7. System displays the due date for the book.

Alternate Flow:

(A1) Book Not Available

1. System displays the message "Book not available for issue."

(A2) Student Reached Issue Limit

1. System displays the message "Maximum book issue limit reached."

Exceptional flow:

- System fails due to database or network error; an error message is displayed.

Termination:

- The system returns to the librarian dashboard.

Post condition:

- The book is successfully issued to the student.
- The database is updated with issue details and due date.

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Lab Experiment - 2

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Functional Requirements

- The system shall allow the librarian to add, update, and delete book records.
- The system shall maintain detailed information for each book including title, author, ISBN, publisher, and category.
- The system shall allow librarians to register, update, and delete library member records.
- The system shall support issuing books to registered members.
- The system shall record book return transactions.
- The system shall automatically update and track book availability status.
- The system shall calculate late return fines based on predefined library rules.
- The system shall allow users to search books by title, author, ISBN, or category.
- The system shall generate reports for issued books, returned books, and overdue books.
- The system shall maintain a complete history of all book transactions.

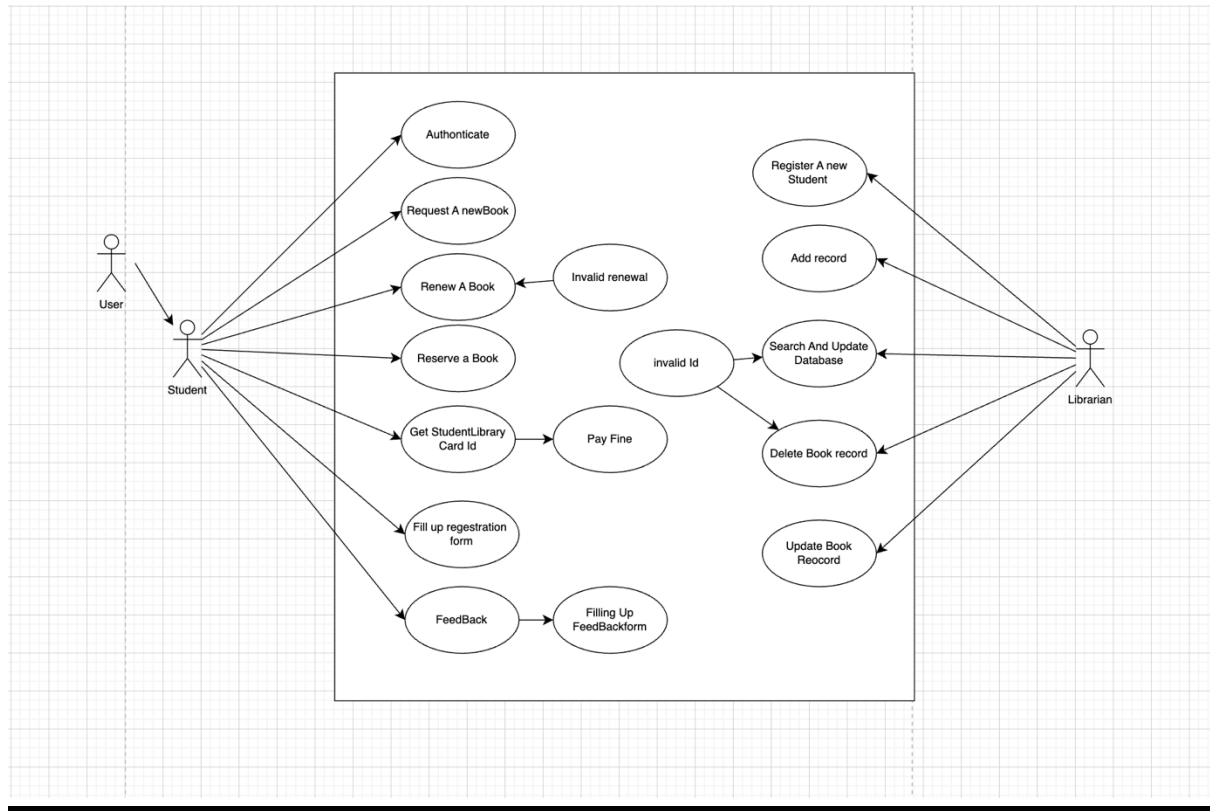
Non-Functional Requirements

- The system shall provide a user-friendly and intuitive interface for all users.
- The system shall ensure data security using role-based access control.
- The system shall respond to user actions within acceptable time limits.
- The system shall be reliable and available during library working hours.
- The system shall support data backup and recovery mechanisms.

User Stories

- As a librarian, I want to add and update book details so that the book catalog remains accurate.
- As a librarian, I want to manage member records so that only authorized users can borrow books.
- As a librarian, I want to issue books to members so that borrowing activities are properly tracked.
- As a librarian, I want to record book returns so that book availability is updated correctly.
- As a librarian, I want the system to calculate fines automatically so that penalties are consistent and fair.
- As a library member, I want to search for books so that I can quickly find required resources.
- As a librarian, I want to view overdue books so that I can follow up with members.
- As a librarian, I want to generate reports so that library usage and performance can be analyzed.
- As a librarian, I want to track book availability so that inventory is managed efficiently.
- As an administrator, I want to control access levels so that system data remains secure.

USECASE DIAGRAM



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Lab Experiment - 3

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Classes in Library Management System

1. User
2. Student
3. Librarian
4. Notification
5. Transaction
6. Book
7. Report

List of methods and attributes for each class:

1. User

Attributes:

- userID : int
- name : String
- email : String
- phone : long int
- password : String

Methods:

- login()
- logout()
- viewProfile()

2. Student

Attributes:

- department : String
- books_borrowed : int

Methods:

- searchBook()

- borrowBook()
- returnBook()
- reserveBook()
- viewFines()
- viewHistory()

3. Librarian

Attributes:

- employeeId : int

Methods:

- addBook()
- updateBook()
- removeBook()
- issueBook()
- approveFine()
- acceptReturn()
- generateReport()

4. Notification

Attributes:

- notificationID : int
- notificationDate : Date
- message : String

Methods:

- sendNotification()

5. Transaction

Attributes:

- transactionID : int
- issueDate : Date
- dueDate : Date
- returnDate : Date
- fineAmount : float

Methods:

- calculateFine() : Float
- recordIssue()

- recordReturn()

6. Book

Attributes:

- bookID : int
- title : String
- author : String
- category : String
- status : boolean
- publisher : String
- isbn : String

Methods:

- checkAvailability()
- updateStatus()
- getDetails()

7. Report

Attributes:

- reportID : int
- reportType : String
- generatedDate : Date

Methods:

- generate()
- export()

Class Responsibility Cards (CRC):

Class: User

Responsibilities:

- Authenticate users
- Maintain common user information

Collaborators:

- Student
- Librarian

Class: Student

Responsibilities:

- Search books
- Borrow, return, and reserve books
- View fines and transaction history

Collaborators:

- Book
- Transaction
- Notification

Class: Librarian

Responsibilities:

- Manage book records
- Issue and accept return of books
- Approve fines
- Generate reports

Collaborators:

- Book
- Transaction
- Report

Class: Book

Responsibilities:

- Store book details
- Track availability status

Collaborators:

- Librarian
- Transaction

Class: Transaction

Responsibilities:

- Record book issue and return details
- Calculate fines
- Maintain borrowing history

Collaborators:

- Student
- Book

Class: Notification

Responsibilities:

- Send notifications for due dates and fines

Collaborators:

- Student

Class: Report

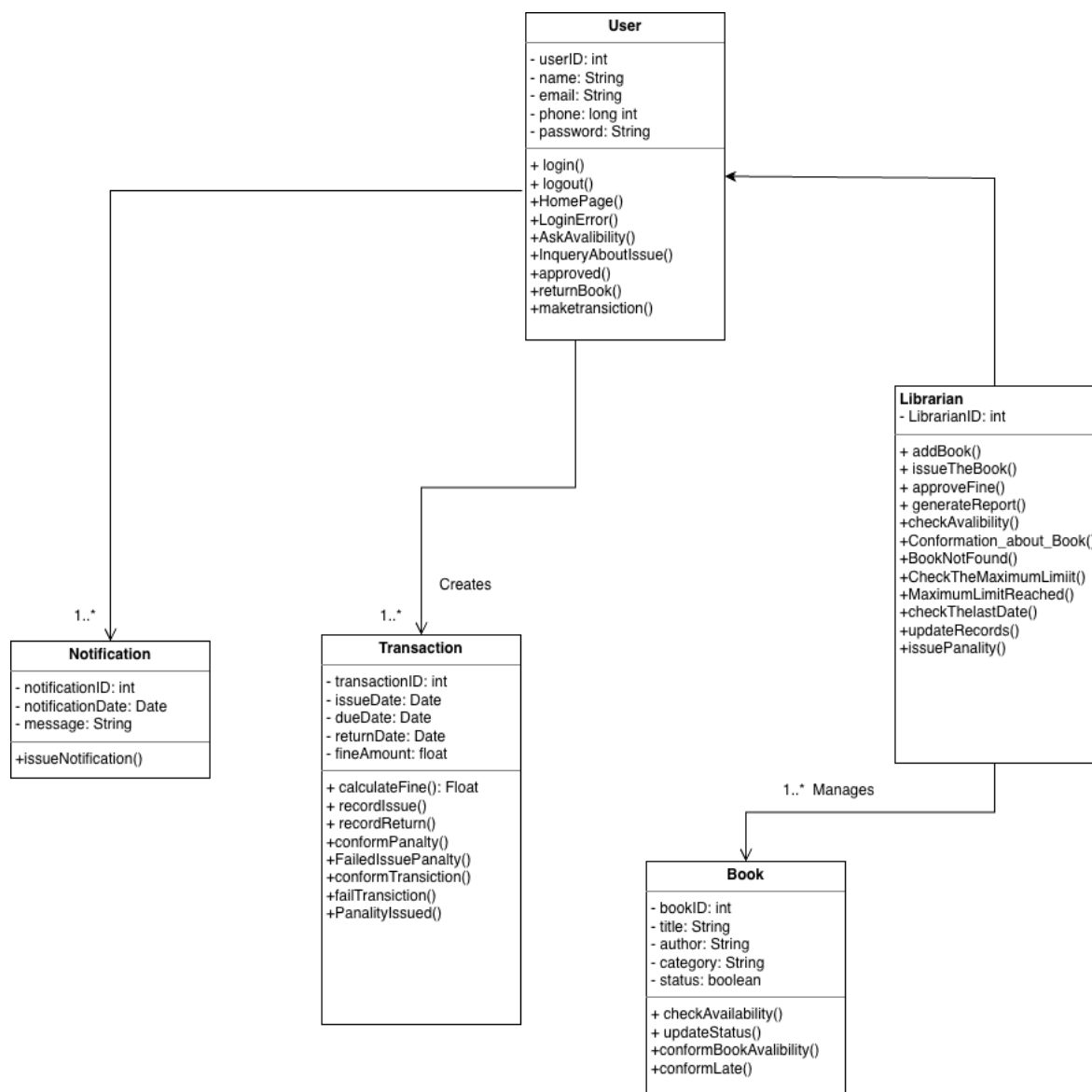
Responsibilities:

- Generate and export library reports

Collaborators:

- Librarian

Class Diagram

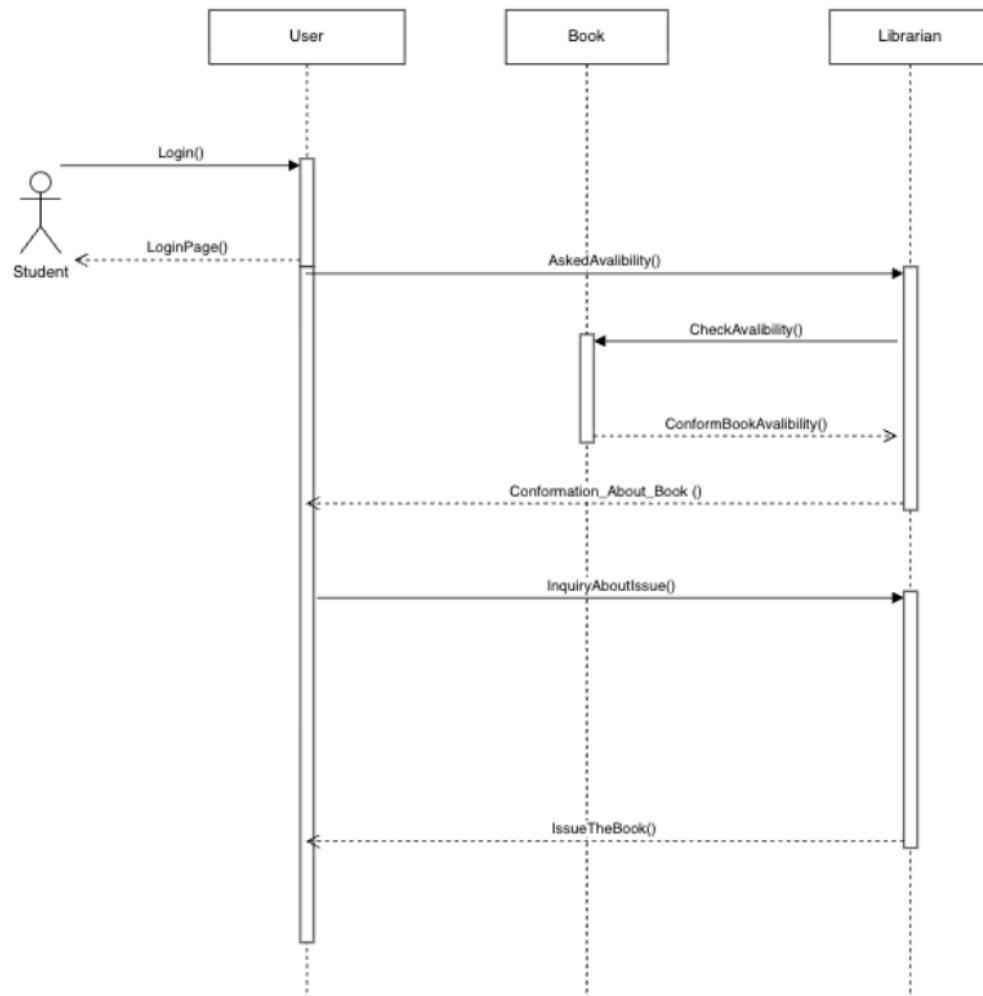


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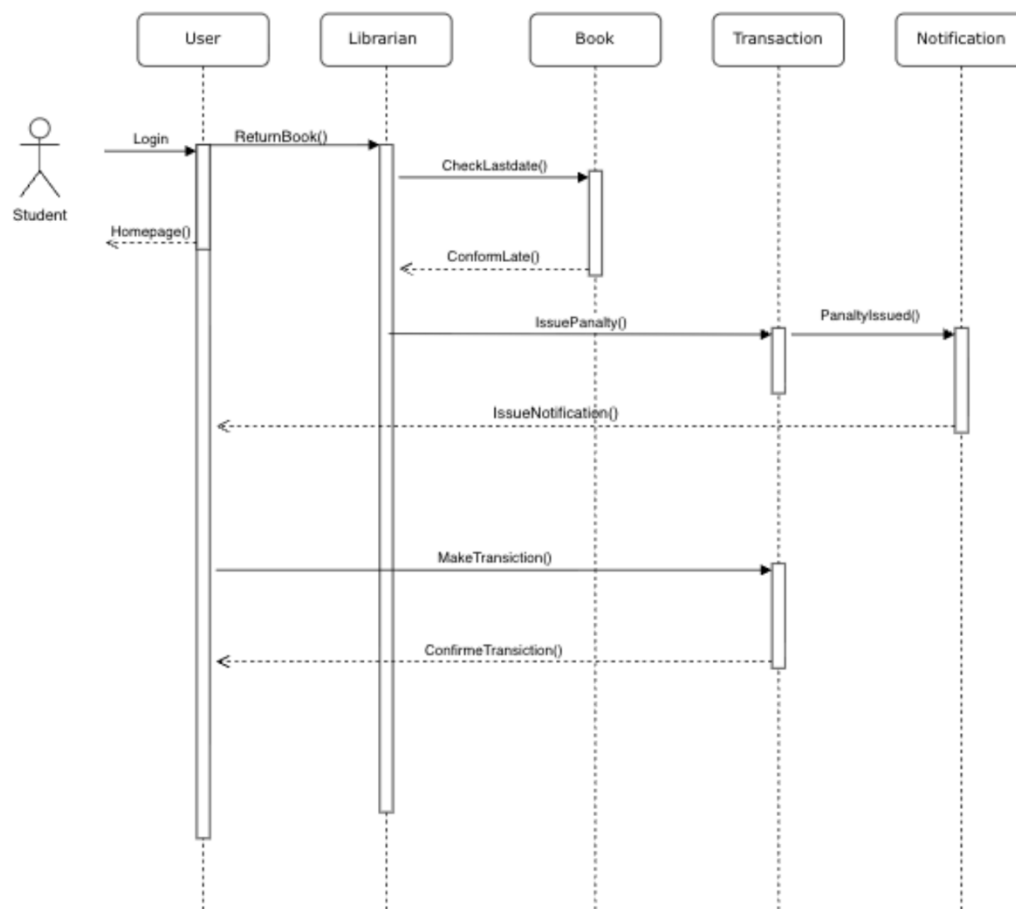
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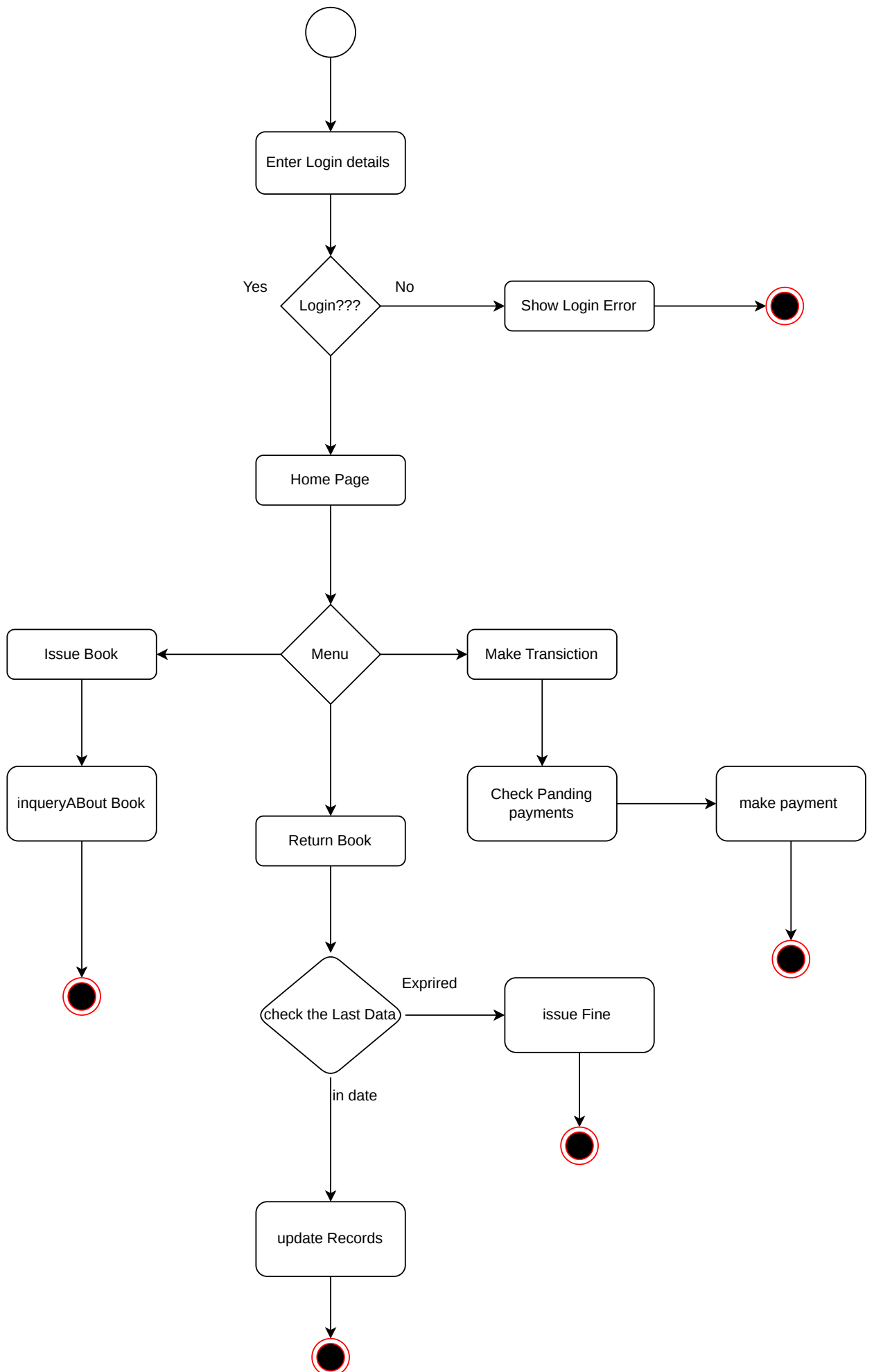
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SequenceDiagram1:



SequenceDiagram2:



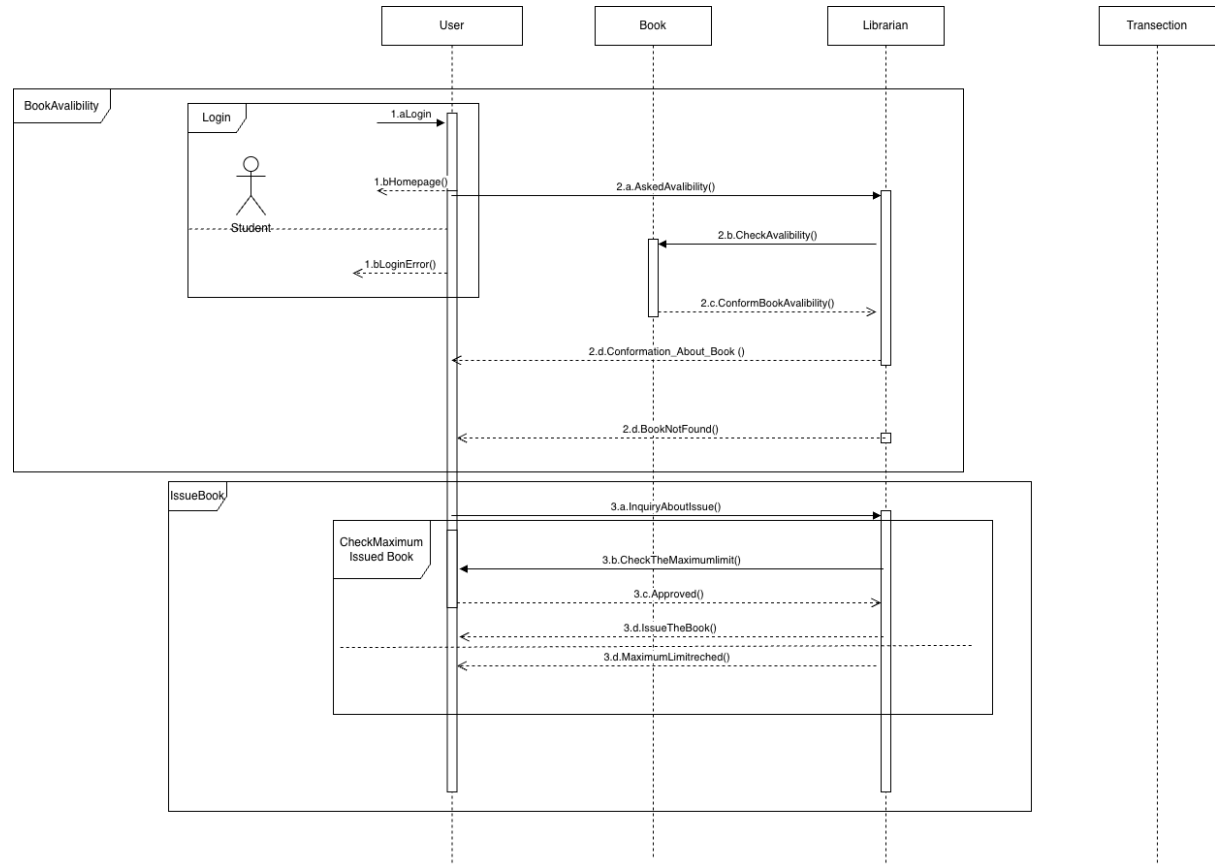


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Lab Experiment - 5

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Interection Diagram-1



Interrection Diagram-2:

X

